HOMEWORK1 REPORT

1.

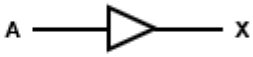
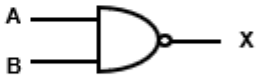
Name	Symbol	Function	Truth Table
AND		X = A • B or X = AB	A B X 0 0 0 0 1 0 1 0 0 1 1 1
OR		X = A + B	A B X 0 0 0 0 1 1 1 0 1 1 1 1
I		X = A'	A X 0 1 1 0
Buffer		X = A	A X 0 0 1 1
NAND		X = (AB)'	A B X 0 0 1 0 1 1 1 0 1 1 1 0
NOR		X = (A + B)'	A B X 0 0 1 0 1 0 1 0 0 1 1 0
XOR Exclusive OR		X = A ⊕ B or X = A'B + AB'	A B X 0 0 0 0 1 1 1 0 1 1 1 0
XNOR Exclusive NOR or Equivalence		X = (A ⊕ B)' or X = A'B'+ AB	A B X 0 0 1 0 1 0 1 0 0 1 1 1

Figure 1. Each logic gates' logic

```
module homework(
        input   wire    a,
        input   wire    b,
        output  wire    and_out,
        output  wire    or_out,
        output  wire    i_out,
        output  wire    buffer_out,
        output  wire    nand_out,
        output  wire    nor_out,
        output  wire    xor_out,
        output  wire    xnor_out
);

        assign and_out  = a && b;
        assign or_out   = a || b;
        assign i_out    = ~a;
        assign buffer_out       = a;
        assign nand_out = ~(a && b);
        assign nor_out  = ~(a || b);
        assign xor_out  = (~a && b) || (a && ~b);
        assign xnor_out = (~a && ~b) || (a && b);

endmodule
```

Figure 2. RTL code

```
module test_bench;

        reg     a,b;
        wire    and_out, or_out, i_out, buffer_out;
        wire    nand_out, nor_out, xor_out, xnor_out;

        homework DUT(.*);

        initial begin
                $dumpfile("wave.vcd");
                $dumpvars(0, test_bench);
                $display("===== Four cases of each logic gates =====");
                $monitor($time, ,"===== With A = %b, B= %b: AND = %b,
OR = %b, I = %b, BUFFER = %b, NAND = %b, NOR = %b, XOR = %b, XNOR = %b
=====",a, b ,and_out, or_out, i_out, buffer_out,nand_out, nor_out,
xor_out, xnor_out);
                a=0; b=0;
                #1;
                a=0; b=1;
                #1;
                a=1; b=0;
                #1;
                a=1; b=1;
                #1;
                $finish;
        end

endmodule
```

Figure 3. Test_bench code

```
C:\Users\ADMIN\Desktop\DSD>vvp sim.out
===== Four cases of each logic gates =====
                 0 ===== With A = 0, B= 0: AND = 0, OR = 0, I = 1, BUFFER = 0, NAND = 1, NOR = 1, XOR = 0, XNOR = 1 =====
                 1 ===== With A = 0, B= 1: AND = 0, OR = 1, I = 1, BUFFER = 0, NAND = 1, NOR = 0, XOR = 1, XNOR = 0 =====
                 2 ===== With A = 1, B= 0: AND = 0, OR = 1, I = 0, BUFFER = 1, NAND = 1, NOR = 0, XOR = 1, XNOR = 0 =====
                 3 ===== With A = 1, B= 1: AND = 1, OR = 1, I = 0, BUFFER = 1, NAND = 0, NOR = 0, XOR = 0, XNOR = 1 =====
test_bench.v:20: $finish called at 4 (1s)

C:\Users\ADMIN\Desktop\DSD>
```

Figure 3. Test_bench result

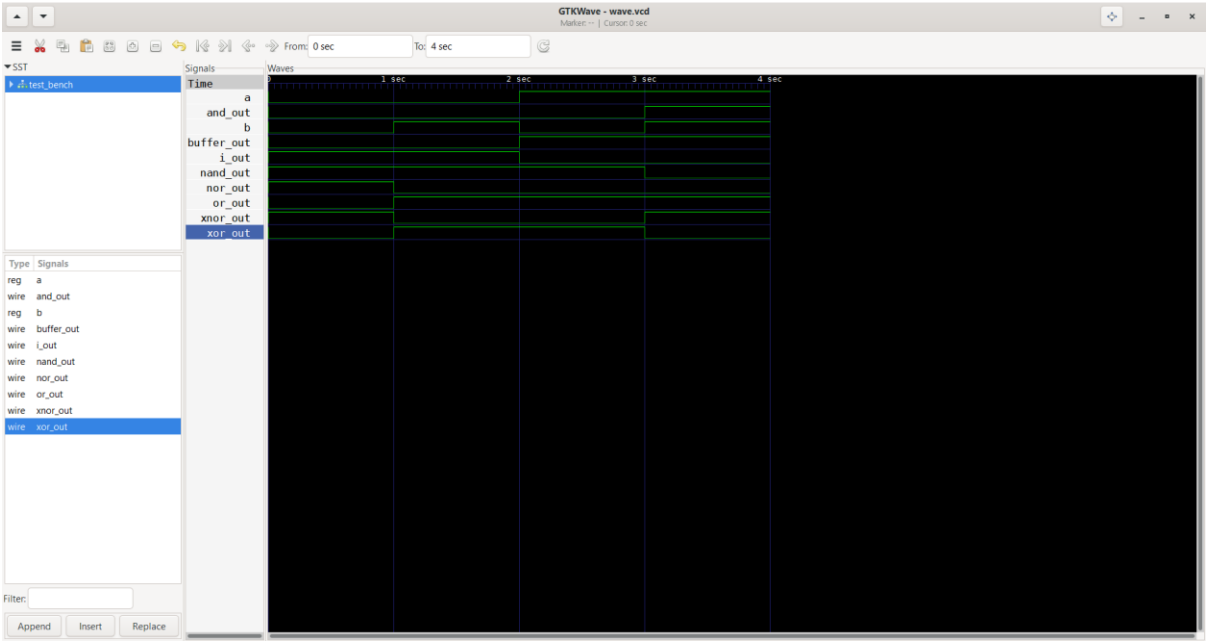


Figure 4. Waveform

2.

(1)

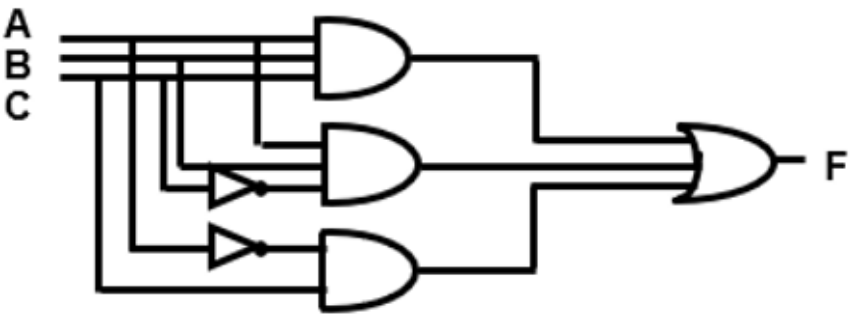


Figure 5. Module 1

```
module first_module(
        input   wire    A,
        input   wire    B,
        input   wire    C,
        output  wire    F
);

        wire    F1, F2, F3;

        assign F1 = A && B && C;
        assign F2 = A && B && ~C;
        assign F3 = ~A && C;

        assign F = F1 || F2 || F3;

endmodule
```

Figure 6. RTL code

```
module test_bench1;

        reg     A, B, C;
        wire    F;

        first_module DUT(.*);

        initial begin
                $dumpfile("wave1.vcd");
                $dumpvars(0, test_bench1);
                $monitor("===== t=%d  A = %b, B = %b, C = %b, F = %b
=====", $time, A, B, C, F);
                A = 0; B = 0; C = 0;
                #1;
                A = 0; B = 0; C = 1;
                #1;
                A = 0; B = 1; C = 0;
                #1;
                A = 0; B = 1; C = 1;
                #1;
                A = 1; B = 0; C = 0;
                #1;
                A = 1; B = 0; C = 1;
                #1;
                A = 1; B = 1; C = 0;
                #1;
                A = 1; B = 1; C = 1;
                #1;
                $finish;
        end

endmodule
```

Figure 7. Test_bench of Module 1

```
C:\Users\ADMIN\Desktop\DSD>vvp sim1.out
VCD info: dumpfile wave1.vcd opened for output.
===== t=                    0  A = 0, B = 0, C = 0, F = 0 =====
===== t=                    1  A = 0, B = 0, C = 1, F = 1 =====
===== t=                    2  A = 0, B = 1, C = 0, F = 0 =====
===== t=                    3  A = 0, B = 1, C = 1, F = 1 =====
===== t=                    4  A = 1, B = 0, C = 0, F = 0 =====
===== t=                    5  A = 1, B = 0, C = 1, F = 0 =====
===== t=                    6  A = 1, B = 1, C = 0, F = 1 =====
===== t=                    7  A = 1, B = 1, C = 1, F = 1 =====
test_bench1.v:28: $finish called at 8 (1s)

C:\Users\ADMIN\Desktop\DSD>
```

Figure 8. Module 1 Result

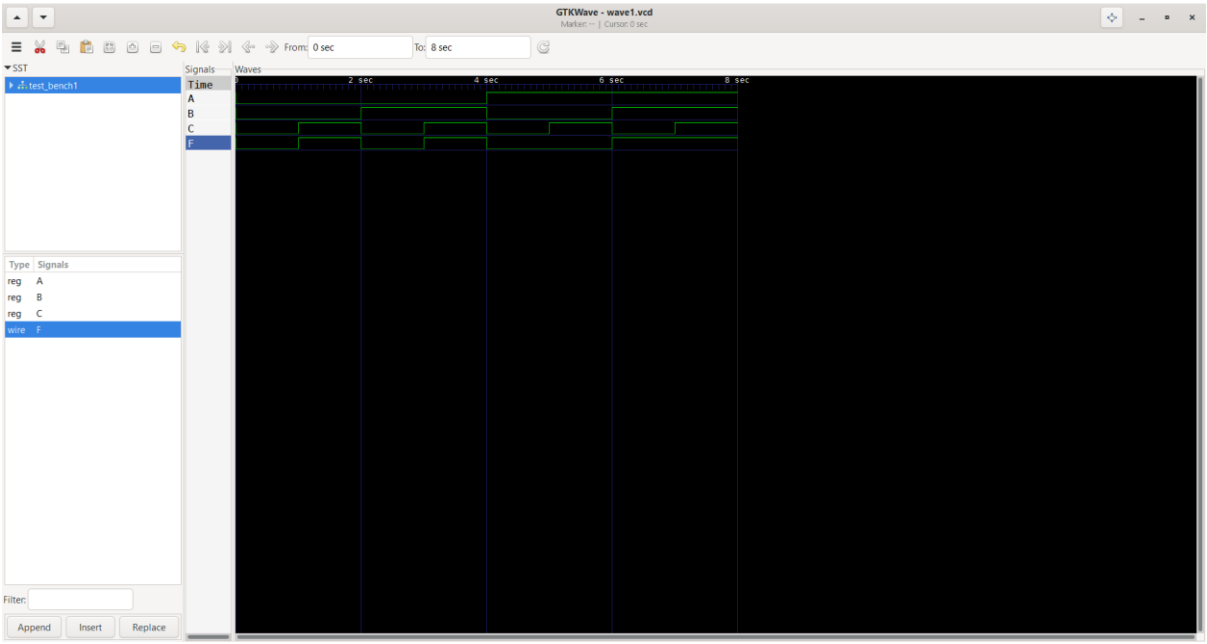


Figure 9. Module 1 Waveform

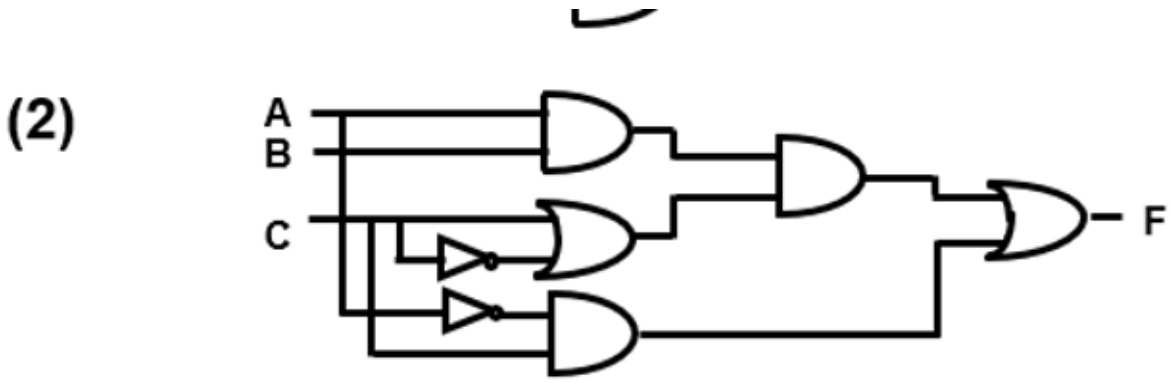


Figure 10. Module 2

```
module second_module(
        input   wire    A,
        input   wire    B,
        input   wire    C,
        output  wire    F
);

        wire    F1, F2;

        assign F1 = (A && B) && (A && ~C);
        assign F2 = ~A && C;

        assign F = F1 || F2 ;

endmodule
```

Figure 11. RTL code

```
module test_bench2;

        reg     A, B, C;
        wire    F;

        second_module DUT(.*);

        initial begin
                $dumpfile("wave2.vcd");
                $dumpvars(0, test_bench2);
                $monitor("===== t=%d  A = %b, B = %b, C = %b, F = %b
=====", $time, A, B, C, F);
                A = 0; B = 0; C = 0;
                #1;
                A = 0; B = 0; C = 1;
                #1;
                A = 0; B = 1; C = 0;
                #1;
                A = 0; B = 1; C = 1;
                #1;
                A = 1; B = 0; C = 0;
                #1;
                A = 1; B = 0; C = 1;
                #1;
                A = 1; B = 1; C = 0;
                #1;
                A = 1; B = 1; C = 1;
                #1;
                $finish;
        end

endmodule
```

Figure 12. Test_bench of Module 2

```
C:\Users\ADMIN\Desktop\DSD>vvp sim2.out
VCD info: dumpfile wave2.vcd opened for output.
===== t=                    0  A = 0, B = 0, C = 0, F = 0 =====
===== t=                    1  A = 0, B = 0, C = 1, F = 1 =====
===== t=                    2  A = 0, B = 1, C = 0, F = 0 =====
===== t=                    3  A = 0, B = 1, C = 1, F = 1 =====
===== t=                    4  A = 1, B = 0, C = 0, F = 0 =====
===== t=                    5  A = 1, B = 0, C = 1, F = 0 =====
===== t=                    6  A = 1, B = 1, C = 0, F = 1 =====
===== t=                    7  A = 1, B = 1, C = 1, F = 0 =====
test_bench2.v:28: $finish called at 8 (1s)

C:\Users\ADMIN\Desktop\DSD>
```

Figure 13. Module 2 Result

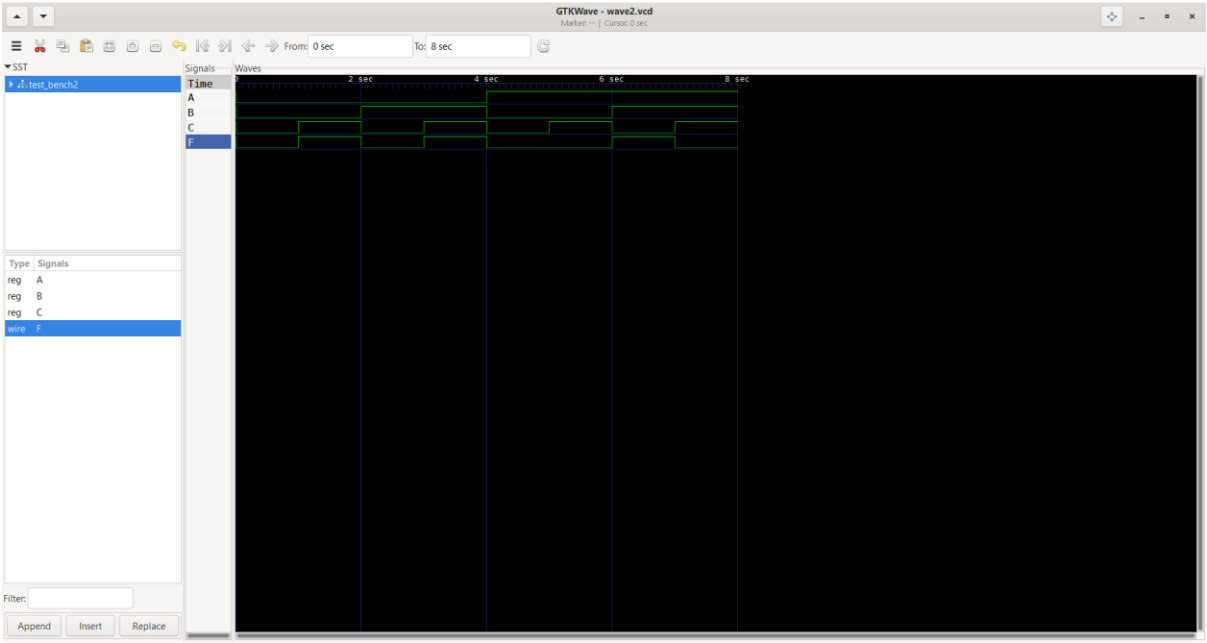


Figure 14. Module 2 Waveform

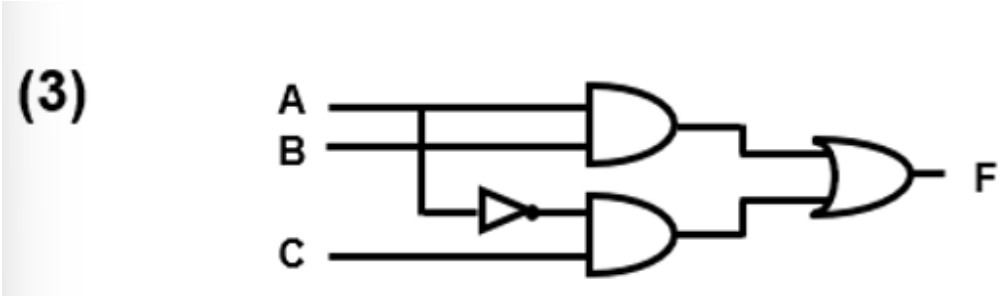


Figure 10. Module 3

```
module third_module(
        input   wire    A,
        input   wire    B,
        input   wire    C,
        output  wire    F
);

        wire    F1, F2;

        assign F1 = A && B;
        assign F2 = ~A && C;

        assign F = F1 || F2 ;

endmodule
```

Figure 11. RTL code

```
module test_bench3;

        reg     A, B, C;
        wire    F;

        third_module DUT(.*);

        initial begin
                $dumpfile("wave3.vcd");
                $dumpvars(0, test_bench3);
                $monitor("===== t=%d  A = %b, B = %b, C = %b, F = %b
=====", $time, A, B, C, F);
                A = 0; B = 0; C = 0;
                #1;
                A = 0; B = 0; C = 1;
                #1;
                A = 0; B = 1; C = 0;
                #1;
                A = 0; B = 1; C = 1;
                #1;
                A = 1; B = 0; C = 0;
                #1;
                A = 1; B = 0; C = 1;
                #1;
                A = 1; B = 1; C = 0;
                #1;
                A = 1; B = 1; C = 1;
                #1;
                $finish;
        end

endmodule
```

Figure 12. Test_bench of Module 3

```
C:\Users\ADMIN\Desktop\DSD>vvp sim3.out
VCD info: dumpfile wave3.vcd opened for output.
===== t=                    0  A = 0, B = 0, C = 0, F = 0 =====
===== t=                    1  A = 0, B = 0, C = 1, F = 1 =====
===== t=                    2  A = 0, B = 1, C = 0, F = 0 =====
===== t=                    3  A = 0, B = 1, C = 1, F = 1 =====
===== t=                    4  A = 1, B = 0, C = 0, F = 0 =====
===== t=                    5  A = 1, B = 0, C = 1, F = 0 =====
===== t=                    6  A = 1, B = 1, C = 0, F = 1 =====
===== t=                    7  A = 1, B = 1, C = 1, F = 1 =====
test_bench3.v:28: $finish called at 8 (1s)
```

Figure 13. Module 3 Result

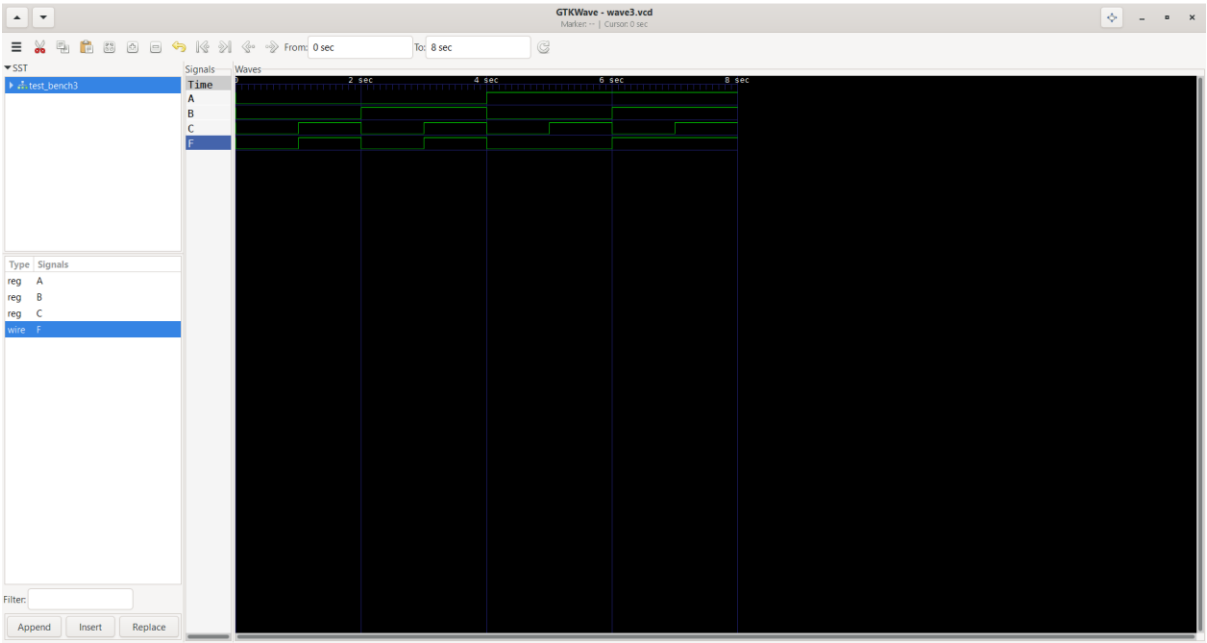


Figure 14. Module 3 Waveform